

Year 7

Year 8

IGCSE

A-Level

I.B

ICT Fun

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Section 4.1: An Introduction to Computer Networking

In this section we will discuss what networks are, including **advantages** and **disadvantages** of using them.

Key Concepts of this section:

- # Understand what a **network** is.
- # Know **what** networks enable us to do.
- # Be able to **describe** advantages and disadvantages setting up a computer network.

Computer networks

What is a computer network?

- # A computer network consists of **2 or more** computers or electronic devices such as printers that have been **linked together** in order to **share data**.
- # Networks can **vary in size**, for example:
 - Home/school networks can consist of only a **few linked** computers
 - Huge networks (**like the internet**) can consist of **billions** of computers.

Computers not connected to a network are known as '**stand-alone**' computers.

What do networks enable us to do?

- # Networks allow users to:
 - Share files
 - Send emails to each other
 - Share resources (**like printers or scanners**).

How are networked computers joined together?

- # Computers on networks are usually connect using:
 - Cables (**if the computers are linked together in the same room/building**)
 - Radio signals (**Wi-Fi**)
 - Telephone lines and satellites (**for long-range connections**).

Advantages and disadvantages of computer networks:

Advantages of computer networks

Connecting computers together on a network offers some advantages:

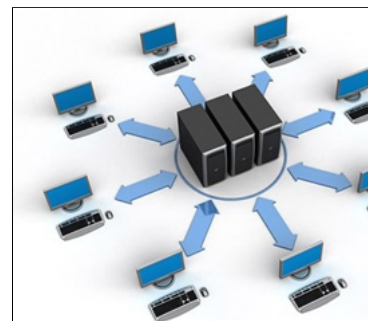
1. Resources such as **printers, scanners** and **internet** can be **shared**
2. Files and data can be **shared**
3. Easy to **communicate** with people on the network (**email, messaging, video-conferencing etc**)



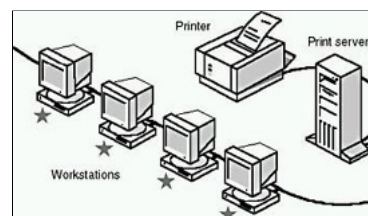
Key Words:

Network, Resources, Link
Communicate.

Examples:



A network is two or more computers joined to



Computers on a network can share the same



Computers can be connected together using network cables.

4. We can access our data from **any computer** (like your student: drive at school)
5. Data can be **backed up** to a separate computer for security
6. Network administrator can **control** the network (changing passwords, monitor the internet etc)

Connecting computers together on the internet offers the following advantages:

1. Access **huge** amount of information and files
2. Communicate with people **around the world** (msn, Skype etc)
3. Access large amounts of media and entertainment (like music, YouTube videos etc)
4. Use on-line services like **online shopping** and **banking** etc

Disadvantages of computer networks

Computers connected to a network can run into problems such as:

1. If **part** of the network **fails** (i.e. a network cable is severed) – this can cause the whole network to **break down** (data can't pass through the damaged cable)
2. Viruses can **spread** across the network very **easily** and **quickly**
3. If a computer on the network is **hacked** into then all other computers are **prone** to attack
4. Networks that use **too few printers** can see printing **queues** develop (due to the amount of people trying to print)

Computers connected over the internet have a different set of disadvantages:

1. We are very prone to **virus** and **spyware infection** from other internet users
2. We are open to '**phishing**' and other **scams** which target internet users for **personal details** such as bank numbers etc
3. We can leave ourselves and our children exposed to **undesirable** material

Activity!

[Intro to Networks - Research Sheet](#)

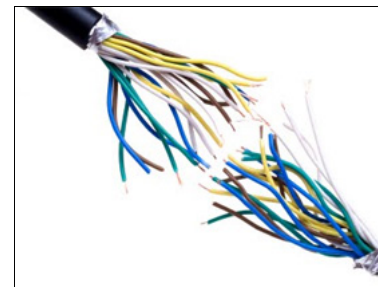
Click the above task and answer the questions about networking basics.



Computers can be connected to a network using Wi-Fi.



School networks allow students to access the internet from any computer.



A network can break down if network cable is damaged.

Document print status					
File	Edit				
Document	Printer	Size	Time Submitted	Status	
(stdin)	ij7	11633k	378792254 days, 13 hours ago	Pending (?)	
(stdin)	ij7	11777k	378792254 days, 13 hours ago	Pending (?)	
(stdin)	ij7	6632k	378792254 days, 13 hours ago	Pending (?)	
(stdin)	ij7	37571k	378792254 days, 13 hours ago	Printing (?)	
(stdin)	ij7	1 pages	Less than one minute ago	Completed	
(stdin)	ij7	1 pages	378792254 days, 13 hours ago	Completed	

Too few printers on a network can cause print queues to develop (printer has too many jobs at once).



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Please add your questions/comments below:

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[Section 4: Networks and the Effects of using them](#)
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Section 18: Data Manipulation

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